

Deep Learning Classification of Skin Lesions Using the ISIC 2019 Challenge Dataset

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Abstract—Accurate classification of skin lesions is critical for early detection of melanoma and other malignancies. Using the ISIC 2019 dataset, we evaluate three pretrained convolutional architectures — EfficientNetB3, ResNet50, and MobileNetV2 — adapted via transfer learning for nine-class lesion classification. We apply systematic ablation and hyperparameter optimization to improve recall on minority classes, then conduct a multimodal fusion study combining CNN image features with an MLP processing patient metadata. We compare all variants on macro F1, recall, and AUC-ROC to identify the most clinically effective approach for automated skin lesion triage. Code is available at <https://github.com/virginia-vrb9e/isic2019-skin-lesion-classification> and https://github.com/Niarfe/DS6050_SPRING_2026_PROJECT. THESE TWO RESPOSITORIES WILL BE FUSED BY PROJECT END.

Index Terms—Skin Lesion Classification, Deep Learning, Convolutional Neural Networks, Transfer Learning, ISIC 2019

I. INTRODUCTION

According to the American Cancer Society, more than 5,000,000 cases of skin cancer are diagnosed annually in the United States [1], [2]. While the vast majority are associated with low mortality rates, new cases of melanoma—the deadliest form of skin cancer—have sharply increased over the past few decades, with an estimated 200,000 new cases expected in 2026 [3]. Yet, even among confirmed cases of melanoma, early detection is associated with a 5-year survival rate of over 95%, creating a dire need for better and more efficient diagnostic tools.

One organization in particular, the International Skin Imaging Collaboration (ISIC), takes a global lead in establishing industry imaging standards while supporting active research. One of their key initiatives is an annual benchmark challenge where teams compete in developing cutting-edge skin lesion classification tools. Inspired by these competitions (but using older data with a full testing set), our motivation and research question will be to explore how we can create high-performing deep learning models able to efficiently classify skin lesions with high sensitivity and specificity.

II. DATASET

The ISIC 2019 aggregate dataset contains 33,569 JPG images and associated metadata compiled from three sources for the ISIC 2019 benchmark challenge. The data is pre-split into training and test sets. Our decision to use the 2019 data,

rather than the most recent ISIC MILK10k, stems from a need to have published test ground truth values so we can perform a comprehensive analysis of our models.

III. LITERATURE REVIEW

A. Toward Errorless Training ImageNet-1k

This paper [6] describes a feedforward artificial neural network trained on the ImageNet 2012 contest dataset to an accuracy rate of 98.3% with a 99.69% Top-1 rate, and an average of 285.9 labels that are perfectly classified over the 10 batch partitions of the dataset. The best performing model uses 322,430,160 parameters with 4 decimal places precision.

In 2023, the author presented a new algorithm called gradient descent tunneling (GDT). The method uses any stochastic gradient descent (SGD) algorithm to train a feedforward neural network (FNN) to a modest accuracy and then follows it up by training the model to zero error rate one data batch at a time, using the idea of homotopy from Numerical Analysis and Dynamical Systems—fields far from Machine Learning (ML).

Gaps Identified: These datasets are relatively small in size. The question is: can GDT be effective for large datasets? An affirmative answer to this question has practical import for predictive precision AI.

B. Advancing Skin Cancer Detection Through Deep Learning and Fusion of Patient Metadata and Skin Lesion Images

Islam et al. [7] describe a deep learning-based tool to address a lack of dermatology practitioners in the UK to provide faster and more accurate triage of skin cancer patients. The central methodology is the concatenation of enhanced metadata with pre-processed images, and the fusion of multiple EfficientNet-B2 models using all or part of the available images and metadata. The best model-fusion combination resulted in 99.50% sensitivity with 82.72% specificity and overall accuracy of 91.11%.

Gaps Identified: (1) The use of a non-public dataset, meaning peer-review of the study's results will be minimal. (2) The volume and lack of diversity in the included patient files, especially the lack of patients with darker skin tones. While melanoma disproportionately affects lighter-skinned individuals, the rates of early detection for darker-skinned individuals lags far behind and would benefit tremendously

from a more robust model design [4]. (3) Artefact removal during image pre-processing was not performed with available libraries, creating a visible reduction in image quality.

C. A Novel Skin Cancer Detection Model Using Modified Finch Deep CNN Classifier

This paper [8] uses a modified Falcon Finch deep convolutional neural network (CNN) while aiming to improve predictive accuracy and computational efficiency. Feature extraction is performed using ResNet-101 deep features, fed into the modified CNN for classification. The dataset consists of 3,297 skin lesion images. Model performance is evaluated with k -fold validation. The results indicate strong performance with accuracy, sensitivity, and specificity values over 90%.

Gaps Identified: First, the dataset is relatively modest in size and the computational demands of combining ResNet-101 features with optimization-based CNN tuning raises questions about hardware requirements and feasibility on larger datasets. Second, the paper has limited discussion about the risk of overfitting—regularization strategies and loss-function analyses are not thoroughly examined. For our project, how can we thoroughly discuss the trade-off between speedy diagnosis and accurate diagnosis with performance metrics?

IV. METHOD

Our initial approach to this research question will be to review the model candidates below and select the highest performing model for further fine-tuning. Our initial candidate models include:

- 1) **AlexNet** – the first large-scale network deployed to beat conventional computer vision methods on a large-scale vision challenge.
- 2) **VGG** – makes use of repeating blocks of elements.
- 3) **NiN (Network in Network)** – convolves whole neural networks patch-wise over inputs.
- 4) **GoogLeNet / Inception v1** – uses networks with multi-branch convolutions.
- 5) **EfficientNet** – a state-of-the-art family of models that have demonstrated improved accuracy and efficiency.
- 6) **ResNet** – one of the most popular off-the-shelf architectures in computer vision.
- 7) **MobileNet** – a family of lightweight models designed for resource-constrained environments. Skin cancer detection models developed with this architecture are well-suited to detection apps that are also an active area of research and development.

We expect that our collective schedules and resource budget will ultimately limit us to two or three choices. Our literature review suggests EfficientNet, ResNet, and MobileNet could provide good outcomes. We also anticipate that as DS 6050 progresses, we could eliminate models from consideration as we learn more about them or identify alternative ones.

A. Pipeline

Our planned workflow includes:

- 1) Examine data and metadata for evidence of imbalances or lack of diversity and join additional sets if needed.
- 2) Establish an image pre-processing pipeline (using either PyTorch’s Torchvision or Albumentations).
- 3) Augment data to avoid overtraining and help the models generalize.
- 4) Build code for selected models, initially focusing on image-only classification and working towards incorporating metadata.
- 5) Select hyperparameters to optimize, load datasets, and execute models using an Overfit-One-Batch Test with the goal that the loss approaches 0.
- 6) Generate training and validation loss and accuracy statistics.

V. EXPERIMENTS: EVALUATION PLAN

After reviewing model candidate statistics, we will down-select appropriate models for our image processing task.

A. Ablation Study

We will evaluate design choices (e.g., batch normalization, residual connections, and channel grouping) to improve our networks by systematically removing or modifying components one-at-a-time to assess the marginal contribution of each. The key question: “Does this component actually help?”

B. Hyperparameter Optimization and Tracking

We will evaluate the usage of Optuna (HPO with pruning), W&B (experiment tracking) and Hydra (configuration management). We will choose a small set of architectures with Hydra, and for each one, tune its training knobs to optimal settings using Optuna.

C. Expected Initial Results

We will compare the best accuracies across models. Table I shows a notional example of the summary of results we expect to produce.

TABLE I
EXAMPLE RESULTS SUMMARY – VALIDATION ACCURACY (NOTIONAL)

Model	Validation Accuracy
ResNet	77.96%
NiN	77.93%
GoogLeNet	77.92%
VGGNet	77.89%
MobileNet	95.17%

D. Evaluation Metrics

The final evaluation will include confusion matrices (predicted vs. actual) with particular attention to:

- **Sensitivity** – correct positive prediction rate.
- **Specificity** – correct negative prediction rate.

VI. DECLARATION ON GENERATIVE AI

During the preparation of this work, the author(s) used Claude (Anthropic) to: create a LaTeX draft using the provided template. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the publication's content.

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